



Mansoura University
Faculty of Computers and Information



Operating Systems

Mass-Storage Systems

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Overview

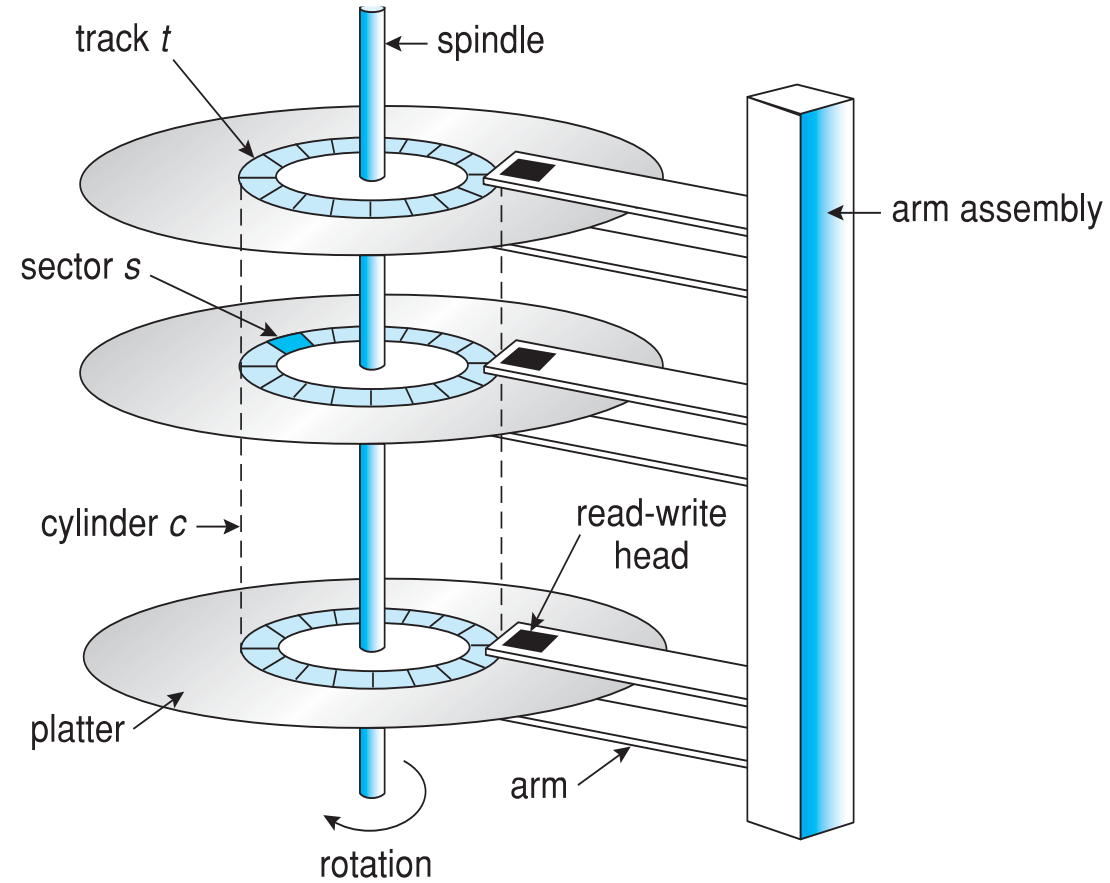
- Magnetic disks are the primary medium for secondary storage.
- Disks may be removable and connect via various buses (e.g., SATA).
- A host controller communicates with disk controller to transfer data.

Disk Structure

Disks are divided into **cylinders**, **tracks**, and **sectors**.

- Each platter surface is divided into concentric tracks.
 - Tracks across platters at the same radius form a **cylinder**.
 - Tracks are split into sectors, which is the smallest unit of data transfer.
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- The OS doesn't deal directly with this geometry.
 - The OS uses a logical block address space (block 0 to block N–1).

Disk Mechanism



Disk Head Positioning

- Two key timing measures affect **positioning time**:
 - **Seek time**: time to position the arm over the correct cylinder (track).
 - **Rotational latency**: time for the desired sector to move under the head.
- **Head crash** occurs if head touches the disk surface (can destroy data).

Disk Performance

- Hard disks vary
 - in capacity (30GB–3TB).
 - in size (commonly 1.8", 2.5", 3.5")
- Performance is measured by:
 - **Transfer Rate** (theoretical): e.g., 6 Gb/s.
 - **Effective Transfer Rate** (actual): e.g., 1 Gb/s.
 - **Seek Time**: ranges from 3–12 ms, with 9 ms being common.
 - **Rotation Latency**: depends on RPM (average latency = half of 60/RPM).

Access Latency and I/O Time

- Access Latency = Average Access Time
- Access Latency = average seek time + average rotational latency
- Data Transfer Time = amount of data to transfer / data transfer rate
- Average I/O Time = Access Latency + Data Transfer Time + Controller Overhead
- **Example:**
 - 4KB transfer at 1Gb/s on a 7200 RPM disk
 - 5ms seek time and 0.1ms controller overhead
- **Answer:**
 - Data transfer time = $4\text{KB} / 1\text{Gb/s} = 0.031\text{ms}$
 - Rotational latency = $\frac{1}{2} (60 / \text{RPM}) = 30 / 7200 = 4.17\text{ms}$
 - Access latency (average access time) = $5\text{ms} + 4.17\text{ms} = 9.17\text{ms}$
 - Therefore, average I/O time = $9.17\text{ms} + 0.031\text{ms} + 0.1\text{ms} = 9.301\text{ms}$

Historical Context

- **IBM Model 350 (1956)**
 - The first commercial disk drive
 - had an access time under 1 second.
 - Held 5 million characters using 50 platters.



Solid-State Disks (SSDs)

- Nonvolatile memory with no moving parts.
- Faster access times than HDDs and more reliable.
- More expensive than HDDs and have shorter lifespans.
- Use buses or connect directly to PCI to bypass bus limitations.

Magnetic Tape

- Sequential storage medium.
- Ideal for backups and archival purposes.
- Compared to disks:
 - Access times are slow (1000x slower).
 - However, transfer rates are comparable once data is located.
- Can store up to 1.5TB, with technologies like LTO and T10000.

Disk Attachment

- Disks connect to computers through I/O buses.
- Commonly used connection technologies include:
 - **USB** (Universal Serial Bus)
 - **SCSI** (Small Computer System Interface)
 - **ATA** (Advanced Technology Attachment)
 - **SATA** (Serial Advanced Technology Attachment)
 - **Fibre Channel (FC)** and **iSCSI** for SAN environments
- Connections influence speed and support for advanced features.
 - E.g., support for command queuing and hot swapping.

Disk Scheduling

- Seek time and rotational delay are costly.
- OS use **disk scheduling** to optimize head movement.
- Several algorithms aim to reduce seek time and improve throughput.

Disk Scheduling Algorithms

- **FCFS (First-Come, First-Served)**

- Processes requests in the order they arrive.
- Simple and fair but can lead to inefficient head movement.

- **SSTF (Shortest Seek Time First)**

- Selects the request closest to the current head position.
- Reduces seek time compared to FCFS but may cause starvation.

Disk Scheduling Algorithms

- **SCAN (Elevator Algorithm)**

- The head moves in one direction then reverses at the end.
- The head serves all pending requests as it passes by them.
- Provides better fairness and average performance than SSTF.

- **C-SCAN (Circular SCAN)**

- Like SCAN but only services requests in one direction.
- After reaching the end the head returns to the beginning.
- The head does not service any requests during the return.
- This provides uniform wait times compared to the SCAN algorithm.

Disk Scheduling Algorithms

- **LOOK**

- A variant of SCAN.
- Avoids unnecessary traversal to the physical end of the disk.
- The head moves only as far as the furthest pending request in each direction.

- **C-LOOK**

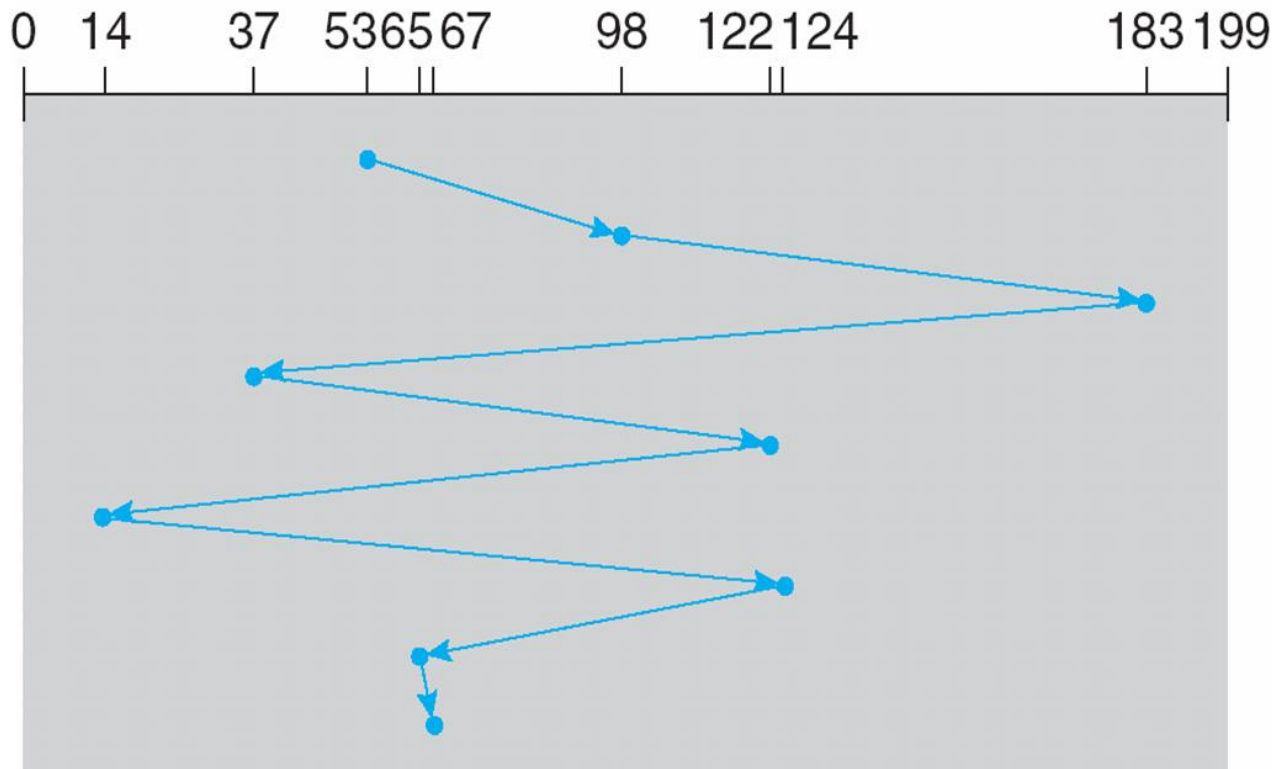
- A variant of C-SCAN.
- Avoids unnecessary traversal to the physical end of the disk.
- Services requests in one direction up to the furthest one, then jumps back.

First-Come, First-Served (FCFS)

Example: Head Movement = 640 Tracks

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53

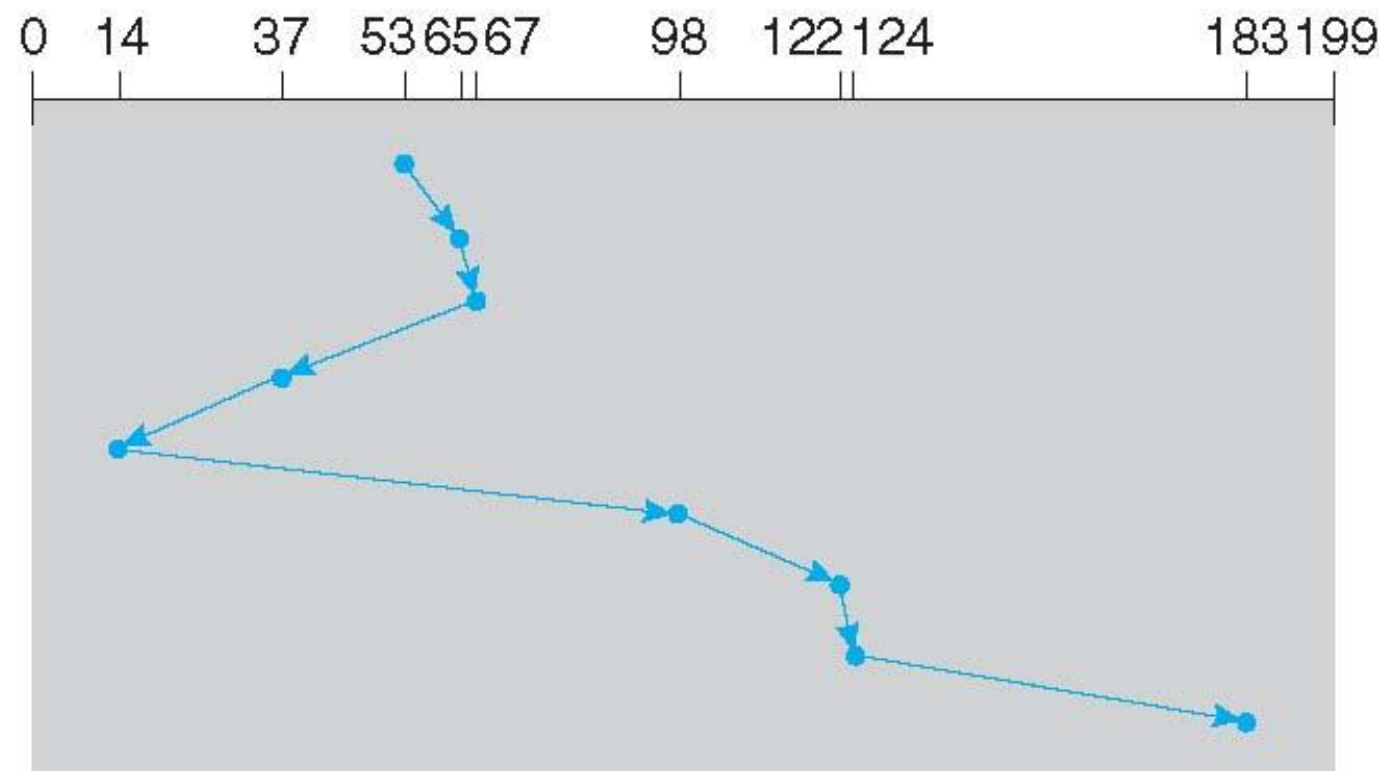


Shortest Seek Time First (SSTF)

Example: Head Movement = 236 Tracks

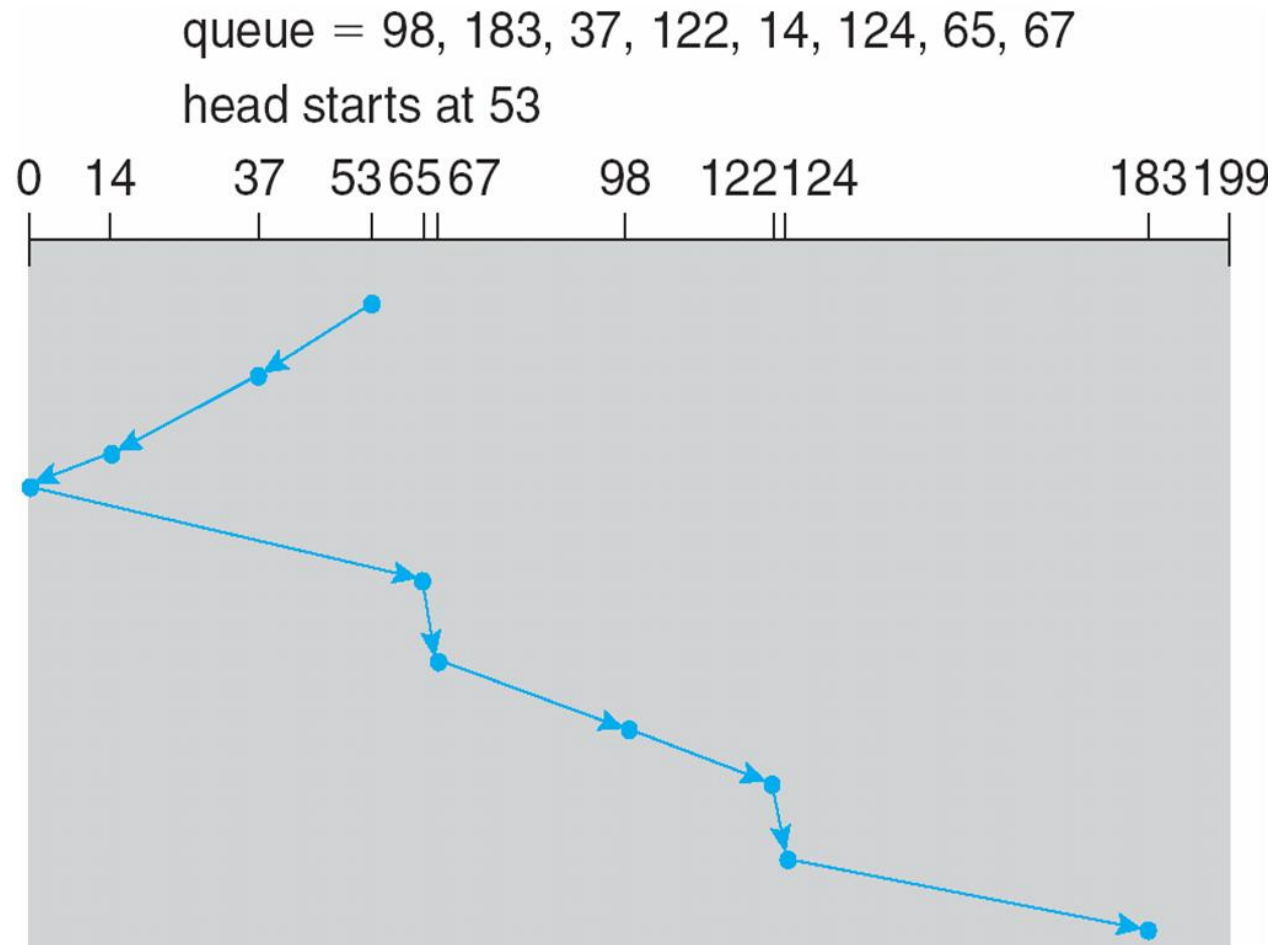
queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53



SCAN

Example: Head Movement = 236 Tracks

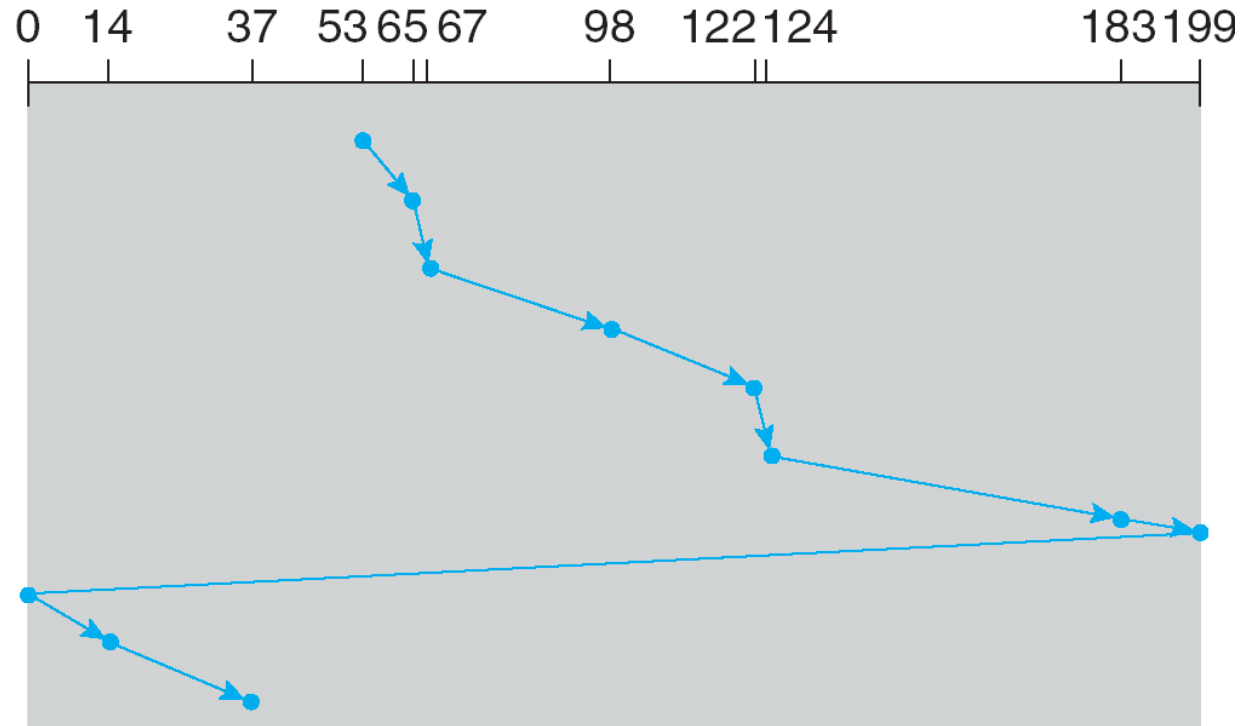


C-SCAN

Example: Head Movement = 183 Tracks

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53

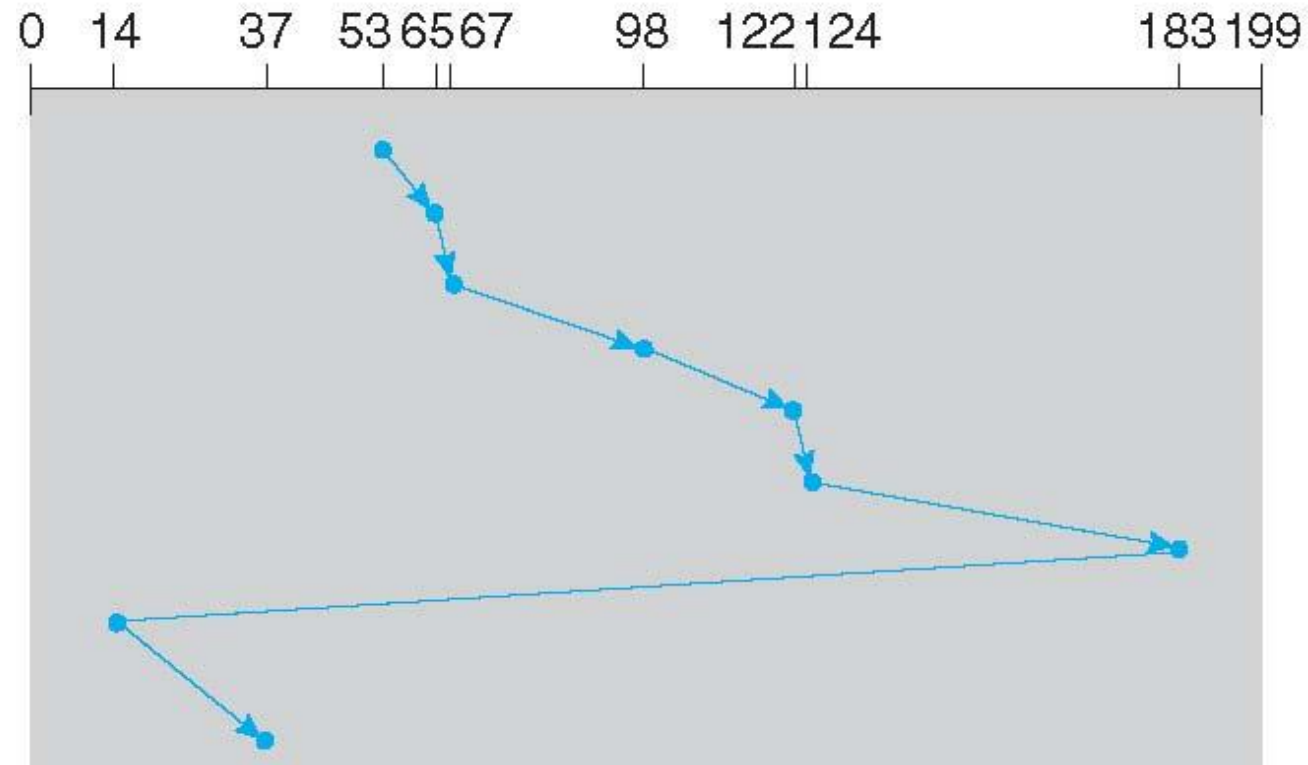


C-Look

Example: Head Movement = 153 Tracks

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53



Disk Scheduling Example

Problem

- Initial head position

53

- Request queue

98, 183, 37, 122, 14, 124, 65, 67

Answer

Algorithm	Number of Tracks
FCFS	640
SSTF	236
SCAN	208
C-SCAN	322
LOOK	184
C-LOOK	236

Choosing a Scheduling Algorithm

- For general-purpose systems, LOOK or C-LOOK are good.
 - They both offer improved performance by minimizing head travel.
- For real-time systems, specialized algorithms may be required.
- Some modern storage (like SSDs) may not use these strategies.

Disk Management

- Disk management includes:
 - **Partitioning** divides the disk into logical sections.
 - **Formatting** writes a file system structure to each partition.
 - **Boot Block** contains operating system loader to initiate startup.
 - **Bad Block Management** detects and marking unusable disk areas.
- Low-level formatting prepares sectors.
- Logical formatting builds the file system.

Swap-Space Management

- **Swap space** is a portion of disk used to extend physical memory.
- When RAM is full, inactive pages are swapped out to free up space.
- **Swap-Space Characteristics:**
 - Traditionally located in a separate partition.
 - Can be a file in the file system (e.g., Windows).
 - Should offer fast access to minimize paging delay.
 - Performance is improved by using a dedicated, contiguous disk region.
- OS may allocate swap space dynamically or statically at boot time.

RAID Structure

- **RAID (Redundant Array of Inexpensive Disks)** can improve:
 - **Performance** via parallelism
 - **Reliability** through redundancy
- RAID groups multiple physical drives into a single logical unit.
- Consequently, it can balance the load and provide fault tolerance.

Thank You